HYPOTHESIS TESTING

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Population Mean: σ KNOWN

15. Individuals filing federal income tax returns prior to March 31 received an average refund of \$1056. Consider the population of "last-minute" filers who mail their tax return during the last five days of the income tax period (typically April 10 to April 15).
    a. A researcher suggests that a reason individuals wait until the last five days is that on average these individuals receive lower refunds than do early filers. Develop appropriate hypotheses such that rejection of H_0 will support the researcher's contention.
    b. For a sample of 400 individuals who filed a tax return between April 10 and 15, the sample mean refund was \$910. Based on prior experience a population standard deviation of $\sigma = \$1600$ may be assumed. What is the p-value?
    c. At $\alpha = .05$, what is your conclusion?
    d. Repeat the preceding hypothesis test using the critical value approach.

16. In a study entitled How Undergraduate Students Use Credit Cards, it was reported that undergraduate students have a mean credit card balance of $3173 (*Sallie Mae*, April 2009). This figure was an all-time high and had increased 44% over the previous five years. Assume that a current study is being conducted to determine if it can be concluded that the mean credit card balance for undergraduate students has continued to increase compared to the April 2009 report. Based on previous studies, use a population standard deviation $\sigma = \$1000$.
    a. State the null and alternative hypotheses.
    b. What is the p-value for a sample of 180 undergraduate students with a sample mean credit card balance of $3325?
    c. Using a .05 level of significance, what is your conclusion?

17. The mean hourly wage for employees in goods-producing industries is currently \$24.57 (Bureau of Labor Statistics website, April, 12, 2012). Suppose we take a sample of employees from the manufacturing industry to see if the mean hourly wage differs from the reported mean of \$24.57 for the goods-producing industries.
    a. State the null and alternative hypotheses we should use to test whether the population mean hourly wage in the manufacturing industry differs from the population mean hourly wage in the goods-producing industries.
    b. Suppose a sample of 30 employees from the manufacturing industry showed a sample mean of \$23.89 per hour. Assume a population standard deviation of \$2.40 per hour and compute the p-value.
    c. With $\alpha = .05$ as the level of significance, what is your conclusion?
    d. Repeat the preceding hypothesis test using the critical value approach.

18. The average annual total return for U.S. Diversified Equity mutual funds from 1999 to 2003 was 4.1% (*BusinessWeek*, January 26, 2004). A researcher would like to conduct a hypothesis test to see whether the returns for mid-cap growth funds over the same period are significantly different from the average for U.S. Diversified Equity funds.

- a. Formulate the hypotheses that can be used to determine whether the mean annual return for mid-cap growth funds differ from the mean for U.S. Diversified Equity funds.
- b. A sample of 40 mid-cap growth funds provides a mean return of $\ddot{x}$= 3.4%. Assume the population standard deviation for mid-cap growth funds is known from previous studies to be σ = *2%*. Use the sample results to compute the test statistic and *p*-value for the hypothesis test.
- c. At α= .05, what is your conclusion?

19. The Internal Revenue Service (IRS) provides a toll-free help line for taxpayers to call in and get answers to questions as they prepare their tax returns. In recent years, the IRS has been inundated with taxpayer calls and has redesigned its phone service as well as posted answers to frequently asked questions on its website (*The Cincinnati Enquirer*, January 7, 2010). According to a report by a taxpayer advocate, callers using the new system can expect to wait on hold for an unreasonably long time of 12 minutes before being able to talk to an IRS employee. Suppose you select a sample of 50 callers after the new phone service has been implemented; the sample results show a mean waiting time of 10 minutes before an IRS employee comes on the line. Based upon data from past years, you decide it is reasonable to assume that the standard deviation of waiting time is 8 minutes. Using your sample results, can you conclude that the actual mean waiting time turned out to be significantly less than the 12-minute claim made by the taxpayer advocate? Use $\alpha = .05$.

22. CCN and ActMedia provided a television channel targeted to individuals waiting in supermarket checkout lines. The channel showed news, short features, and advertisements. The length of the program was based on the assumption that the population mean time a shopper stands in a supermarket checkout line is 8 minutes. A sample of actual waiting times will be used to test this assumption and determine whether actual mean waiting time differs from this standard.

a. Formulate the hypotheses for this application.
b. A sample of 120 shoppers showed a sample mean waiting time of 8.4 minutes. Assume a population standard deviation of $\sigma = 3.2$ minutes. What is the p-value?
c. At $\alpha = .05$, what is your conclusion?
d. Compute a 95% confidence interval for the population mean. Does it support your conclusion?

Population Mean: σ UNKNOWN

27. The Employment and Training Administration reported that the U.S. mean unemployment insurance benefit was $238 per week (*The World Almanac*, 2003). A researcher in the state of Virginia anticipated that sample data would show evidence that the mean weekly unemployment insurance benefit in Virginia was below the national average.
    a. Develop appropriate hypotheses such that rejection of H_0 will support the researcher's contention.
    b. For a sample of 100 individuals, the sample mean weekly unemployment insurance benefit was $231 with a sample standard deviation of $80. What is the p-value?
    c. At $\alpha = .05$, what is your conclusion?
    d. Repeat the preceding hypothesis test using the critical value approach.

28. A shareholders' group, in lodging a protest, claimed that the mean tenure for a chief executive office (CEO) was at least nine years. A survey of companies reported in *The Wall Street Journal* found a sample mean tenure of $\bar{x} = 7.27$ years for CEOs with a standard deviation of $s = 6.38$ years (*The Wall Street Journal*, January 2, 2007).

a. Formulate hypotheses that can be used to challenge the validity of the claim made by the shareholders' group.

b. Assume 85 companies were included in the sample. What is the p-value for your hypothesis test?

c. At $\alpha = .01$, what is your conclusion?

31. The Coca-Cola Company reported that the mean per capita annual sales of its beverages in the United States was 423 eight-ounce servings (Coca-Cola Company website, February 3, 2009). Suppose you are curious whether the consumption of Coca-Cola beverages is higher in Atlanta, Georgia, the location of Coca-Cola's corporate headquarters. A sample of 36 individuals from the Atlanta area showed a sample mean annual consumption of 460.4 eight-ounce servings with a standard deviation of $s = 101.9$ ounces. Using $\alpha = .05$, do the sample results support the conclusion that mean annual consumption of Coca-Cola beverage products is higher in Atlanta?

33. Annual per capita consumption of milk is 21.6 gallons (*Statistical Abstract of the United States: 2006*). Being from the Midwest, you believe milk consumption is higher there and wish to support your opinion. A sample of 16 individuals from the midwestern town of Webster City showed a sample mean annual consumption of 24.1 gallons with a standard deviation of $s = 4.8$.

a. Develop a hypothesis test that can be used to determine whether the mean annual consumption in Webster City is higher than the national mean.

b. What is a point estimate of the difference between mean annual consumption in Webster City and the national mean?

c. At $\alpha = .05$, test for a significant difference. What is your conclusion?